

Reviving, reproducing, and revisiting Axelrod’s second tournament

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1 Introduction

The Prisoner’s Dilemma (PD) has extensively studied as a model of cooperative behaviour since Robert Axelrod’s seminal work in the 1980s [1, 2, 4]. Robert Axelrod invited fellow academics and the wider public, through computer hobbyist magazines, to write computer code to play the Iterated Prisoner’s Dilemma (IPD). This initial research has subsequently led to a huge area of research aiming to understand the emergence of cooperative behaviour. A good overview of the variety of research areas is given in [14].

In the Prisoner’s Dilemma, each player chooses simultaneously and independently between cooperation (C) or defection (D). The payoffs of the game are defined by the matrix $\begin{pmatrix} R & S \\ T & P \end{pmatrix}$, where $T > R > P > S$ and $2R > T + S$. The PD is a one round game and when studied as a repeated game is called the Iterated Prisoner’s Dilemma. In Robert Axelrod’s tournaments the particular values $R = 3, P = 1, S = 0, T = 5$ were used.

Many strategies for playing the IPD have been developed in the preceeding 50 years, often defined by a bit of source code yet described insufficiently in prose or with insufficient data for reverse engineering when published, and often without the original source code. This makes reproducing earlier results difficult or impossible and sometimes leads new researchers to test their strategies against small collections of easily implemented strategies, or to re-use an existing small collection of strategies. These strategy sets can be biased toward simple or cooperative strategies and otherwise lack diversity, which makes claims of the performance of new strategies less robust. Moreover, new strategies are often declared to be the new best strategy for the IPD but are rarely tested against former supposed champions.

In Robert Axelrod’s famous tournaments, strategies were submitted written in either the Fortran [16] or Basic [5] programming languages (in the later case they were manually translated to Fortran). The very first tournament [1] involved 14 submissions (complemented by a 15th random strategy). Famously, the winner of this tournament was Tit For Tat (TFT), a strategy that mimics the opponent’s previous action. The only sources for this work are the paper itself, with all of the source code apparently lost. In the second tournament [2] 62 strategies were submitted as well as a further random strategy and again TFT was declared the winner. This tournament is better preserved from a computation archaeological point of view as a version of the source code for all the strategies was kept and made available for use. Code used to test the various strategies and tournament structure still exists and was made available by Robert Axelrod.¹ However the final source code used to run the tournaments reported in [2] is apparently no longer extant.

This manuscript describes the process of reviving and using these strategies in a modern software framework ([18]): a Python library with over 200 strategies and a large number of analytical procedures which has been used in ongoing research [11, 12, 9, 6, 13].

Importantly, reviving the strategies is not done through a manual exercise of reverse engineering the Fortran code which would be prone to mistakes. This is done by calling the original functions ensuring a **high-fidelity reproduction and analysis of Robert Axelrod’s original work**. This will be described in more detail in Section 2. Results and further analyses pertaining to reproducing the tournament will be given in Section 3. Finally, Section 4 will extend the analysis to include contemporary research:

- Experimenting with adding one, two or three, of over 200 other strategies from [18] to see how the results change.
- Running a tournament with all considered strategies (more than 250).

This work contributes to the game theoretic literature by providing the first exercise in reproducing reported results that have been at the core of the study of cooperation. Furthermore it also provides a contemporary lens which, amongst other things concludes with one of the largest ever IPD tournaments. Finally, by reviving the original code and making it available to use in [18] we preserve for future use the original strategies of Robert Axelrod’s second tournament.

¹This source code can be found at <http://www-personal.umich.edu/~axe/research/Software/CC/CC2/TourExec1.1.f.html>

2 Reviving the tournament

As described in Section 1, the original source code for Axelrod’s second tournament was written in Fortran, subsequently published at [3]. This website maintained by the University of Michigan Center for the Study of Complex Systems was last updated (at the time of writing) in 1996. The source code was originally a single file (TourExec1.1.f) and is published on the site in HTML format.

For the purposes of this work, the html formatting was removed to produce the original fortran file which was then minimally modified so that it would compile on a modern compiler. Furthermore, each strategy was extracted into a single modular file for ease of examination and compilation. Finally, a Makefile was created to control the compilation of the fortran strategy files into a single binary shared object file (named `libstrategies.so`) which is then installed into a standard location on a Posix compliant operating system. This shared library provides runtime access to the original code. The modified code is available at <https://github.com/Axelrod-Python/TourExec> and has been archived at [7].

A Python library has been written that enables an interface between the Fortran code and the Axelrod library described in the previous section. This library is referred to as the `Axelrod_fortran` library and is available at <https://github.com/Axelrod-Python/axelrod-fortran>. The specific version used for this work is [8]. This library has the binary file `libstrategies.so` described above as a dependency and offers a straightforward to install and use option for the study of the strategies of [2] (using standard scientific Python packages).

For this to work there are a variety of minor translations that need to take place. As documented at <http://www-personal.umich.edu/~axe/research/Software/CC/CC2/TourExec1.1.f.html> the Fortran code implies that each strategy is a Fortran function which takes the following inputs.

- J: The opponent’s previous move: 0 corresponds to a defection and 1 to a cooperation.
- M: The current turn number (starting at 1).
- K: The player’s current cumulative score.
- L: The opponent’s current cumulative score.
- R: A random number between 0 and 1: used by stochastic strategies.
- JA: The player’s previous move.

For example, Figure 1 shows the source code for `k92r.f` also known as Tit For Tat.

```
FUNCTION K92R(J,M,K,L,R, JA)
C BY ANATOL RAPOPORT
C TYPED BY AX 3/27/79 (SAME AS ROUND ONE TIT FOR TAT)
c replaced by actual code, Ax 7/27/93
c T=0
c K92R=ITFTR(J,M,K,L,T,R)
    k92r=0
    k92r = j
c test 7/30
c write(6,77) j, k92r
c77 format('␣test␣k92r.␣j,k92r:␣', 2i3)
RETURN
END
```

Figure 1: Fortran Source code for original Tit For Tat strategy submitted to Axelrod’s second tournament.

The `Axelrod` library takes advantage of the modern Object Oriented framework in Python. Each strategy is a class with agent-based behaviour. The input to each player’s strategy function is simply the opponent’s class, providing access to both players history of plays. In the newly written `Axelrod_fortran` library a class inherited from the base `Axelrod` class is written that interfaces with the Fortran strategies and the Python requirements. This includes, for example passing an initial move required by the Fortran player. Using the original Fortran code as reference (see Figure 2) it is assumed that the assumed prior move is a cooperation (which is in line with Figure 1).

```

67      Do 20 Game = 1,5
68          RowGameSc = 0
69          ColGameSc = 0
70          JA = 0          ! Row's previous move, reported to column
71          JB = 0          ! Col's previous move, reported to row

```

Figure 2: A portion of the code <https://github.com/Axelrod-Python/TourExec/blob/master/src/tournament/AxTest.f> setting the default previous move to a cooperation and score to 0.

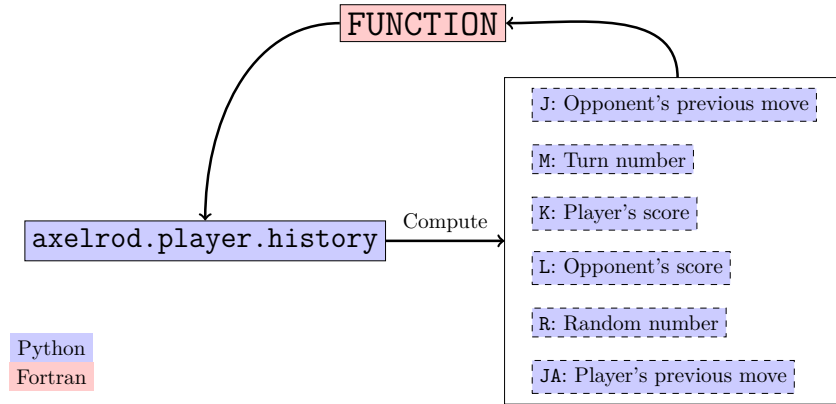


Figure 3: The interface between the Python axelrod library and the Fortran code

Figure 3 shows a diagrammatic representation of the interface between the Python strategy and the Fortran function. The major advantage of this approach is that at no point has any subjectivity been added to the process of reproducing Axelrod’s second tournament. Indeed for some strategies the only description available is the Fortran code itself, thus they are being run and used as originally coded. To the authors’ knowledge this is the best known way to reproducing Axelrod’s work.

3 Reproducing Axelrod’s second tournament

From [2], the following characteristics of the original tournament have been identified:

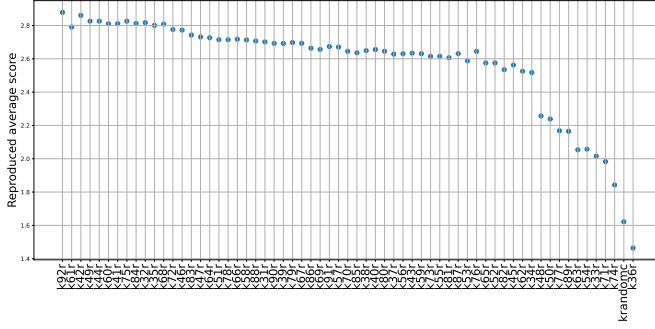
- Matches have length from {63, 77, 151, 308, 156};
- Players do not know the number of rounds in a given match;
- A total of 25,000 repetitions across the various match lengths have been carried out.

Note that there is some lack of clarity in [2] as to the length of the matches:

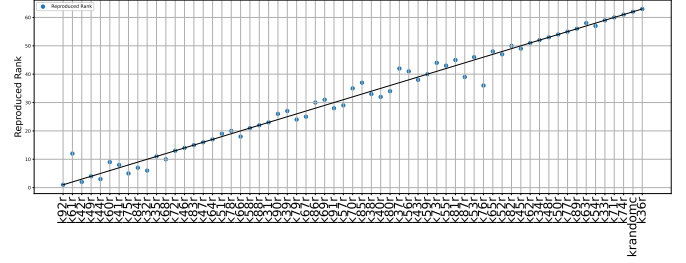
“As announced in the rules, the length of the games was determined probabilistically with a .00346 chance of ending with each given move. This parameter was chosen so that the expected median length of a game would be 200 moves. In practice, each pair of players was matched five times, and the lengths of these five games were determined once and for all by drawing a random sample. The resulting random sample from the implied distribution specified that the five games for each pair of players would be of lengths 63, 77, 151, and 308 moves. Thus the average length of a game turned out to be somewhat shorter than expected at 151 moves.”

As only four match length samples were specified but the average was given, a fifth match length of 156 is assumed (giving the correct average of 151). It seems that the only stochastic smoothing used was these five repetitions. Without a known random seed it is not possible to replicate the original tournament exactly, thus the tournament is repeated a total of 25,000 for each match length.

The replicated scores and corresponding rankings of each strategy across all repetitions are shown in Figure 4.



(a) Average score per turn of each strategy.



(b) Ranking of each strategy.

Figure 4: Reproduced tournament with strategies ordered by original rank

The top 15 strategies in the reproduced tournament are shown in Table 1. Whilst the results show an overall agreement with the original reported results, a distinct outlier is strategy **k61r**, referred to as Champion, after the last name of its author. Champion cooperates for the first 10 moves, plays TFT for the next fifteen and then will cooperate unless: the other player defected on the previous move, the other player cooperated less than 60% and a random number between 0 and 1 is greater than the other player's cooperation rate. Upon closer investigation a bug was noted in the Fortran code for Champion. One initial value was not being initialised, the modified version of this strategy is shown in Figure 5

	Author	Scores	Reproduced Rank	Original Rank	Mean Cooperation Rate
k92r	Anatol Rapoport	2.878	1	1	0.922
k61r	Danny C Champion	2.791	12	2	0.954
k42r	Otto Borufsen	2.861	2	3	0.916
k49r	Rob Cave	2.826	4	4	0.892
k44r	William Adams	2.826	3	5	0.882
k60r	Jim Graaskamp and Ken Katzen	2.810	9	6	0.844
k41r	Herb Weiner	2.811	8	7	0.865
k75r	Paul D Harrington	2.826	5	8	0.802
k84r	T Nicolaus Tideman and Paula Chieruzz	2.812	7	9	0.882
k32r	Charles Kluepfel	2.817	6	10	0.873
k35r	Abraham Getzler	2.801	11	11	0.888
k68r	Fransois Leyvraz	2.808	10	12	0.917
k72r	Edward C White Jr	2.776	13	13	0.924
k46r	Graham J Eatherley	2.772	14	14	0.948
k83r	Paul E Black	2.743	15	15	0.935

Table 1: Top 15 strategies in the reproduced tournament

It is not clear if this bug affected the tournament results reported in 1980. There is no source of the specific code used to run the tournaments and the bug only effects Champion on a second use of the function (the very first time: IC00P is assumed to be 0). Thus, it is possible that every single match of the tournament was run in isolation.

Despite fixing this apparent bug and verifying all other strategies for potentially similar bugs there is still a discrepancy in the results. Figure 6 shows the results of the tournament computed using a number of different approaches.

There is no immediate explanation for the remaining discrepancy with the results reported in [2]. We note that against a primarily cooperative opponent, Champion can play similarly to TFT, which would explain its similar performance. Potential explanations include:

- An unknown modification of the source code of Champion (**k61r**) used in the final original tournament code [2]
- Stochastic variation not being sufficiently taken in to account in the original tournament [2].
- A difference with how an older Fortran compiler would interpret the commands, though the implemented version seems to interact as expected.

```

1      FUNCTION K61R(ISPICK,ITURN,K,L,R, JA)
2  C BY DANNY C. CHAMPION
3  C TYPED BY JM 3/27/79
4      k61r=ja      ! Added 7/27/93 to report own old value
5      IF (ITURN .EQ. 1) ICOOP = 0      ! Added 10/8/2017 to fix bug for multiple runs
6      IF (ITURN .EQ. 1) K61R = 0
7      IF (ISPICK .EQ. 0) ICOOP = ICOOP + 1
8      IF (ITURN .LE. 10) RETURN
9      K61R = ISPICK
10     IF (ITURN .LE. 25) RETURN
11     K61R = 0
12     COPRAT = FLOAT(ICOOP) / FLOAT(ITURN)
13     IF (ISPICK .EQ. 1 .AND. COPRAT .LT. .6 .AND. R .GT. COPRAT)
14 +K61R = 1
15     RETURN
16     END

```

Figure 5: Original code for k61r with fixed bug on line 5.

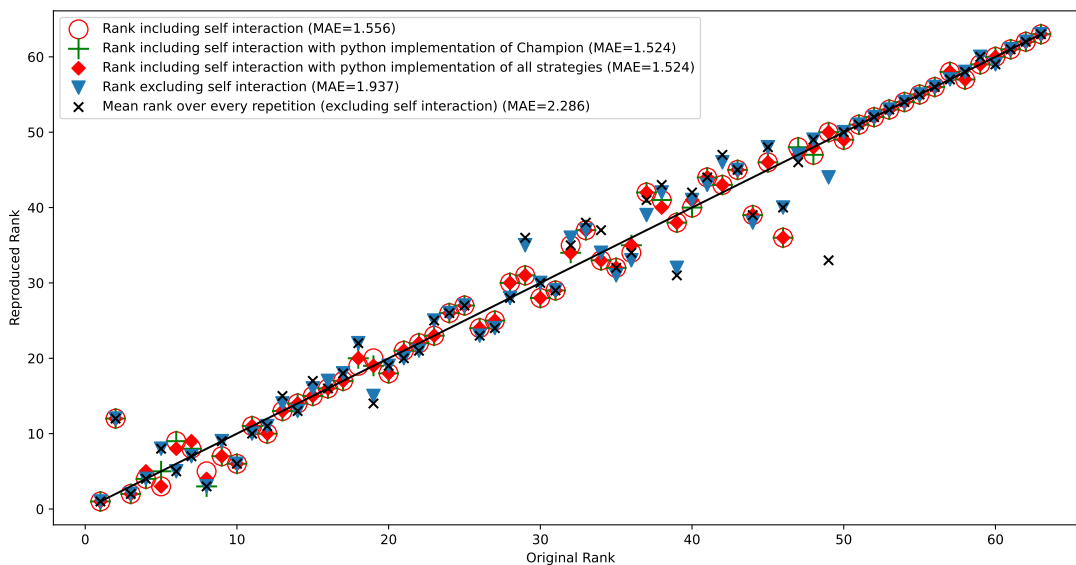


Figure 6: Using a variety of approaches to reproduce the original tournament. This includes using a Python implementation of Champion's strategy as well as using as many Python implementation of the Fortran strategies as possible.

Apart from the strategy by Champion, the agreement between the original and the reproduced tournament is quite close. The main conclusions included for example that TFT (k92r) once again wins the tournament. Furthermore, the fact that high performing strategies are “nice” is also evident although perhaps interestingly k42r which takes the second rank of 61r is a strategy that cooperates with most strategies apart from itself. The overall cooperation rate of the tournament is 0.750 . Figure 7 shows the cooperation rates of the tournament. It is clear that the high performing strategies cooperate overall more often. Looking at the pairwise cooperation rates in Figure 7b shows that the high performing strategies generally seems to cooperate with high performing strategies. This underpins one of the main conclusions of [2] explaining the emergence of cooperation in competitive environments.

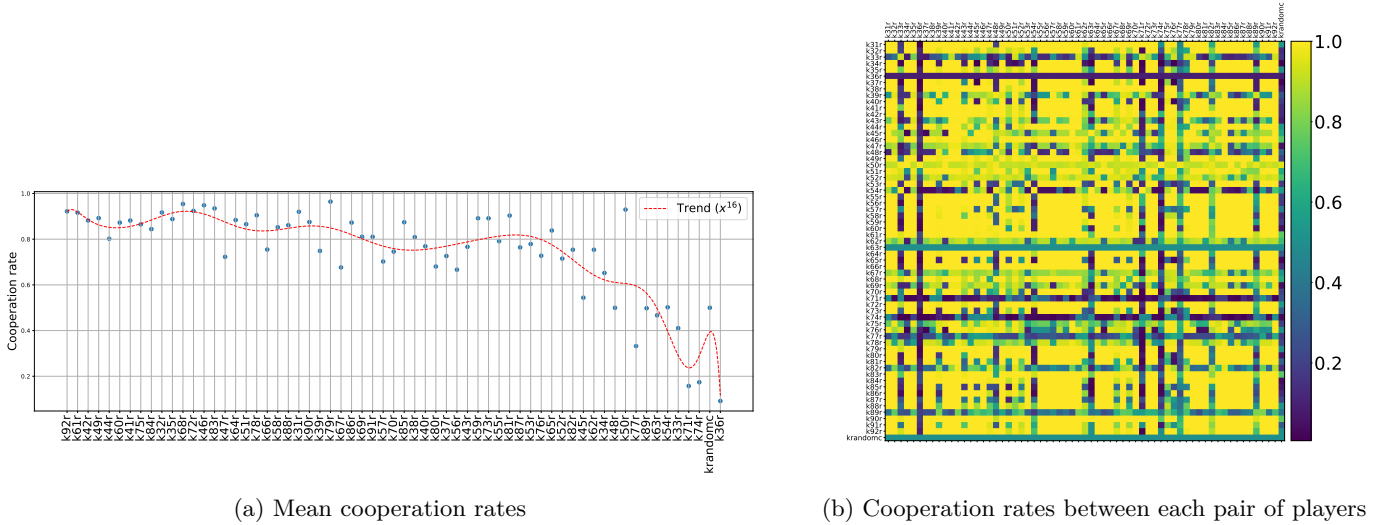


Figure 7: Replicated tournament cooperation rates with strategies ordered by original rank

In [2], a linear regression model is used to identify 5 strategies, the scores against which are good predictors of the overall performance. The reported R^2 value is 0.979 (indicating 97% of variance accounted for by the model). For completeness, the coefficients of this model (reported in [2]) are shown in Table 2.

Strategies	Coefficients
k69r	0.202000
k91r	0.198000
k40r	0.110000
k76r	0.072000
k67r	0.086000
Intercept	0.795000

Table 2: Linear model described in [2] with $R^2 = 0.7589$

Given the discrepancy in results shown in Figure 4 and Table 1 it is not surprising to see that this model no longer performs as well with $R^2 = 0.7589$.

Fitting a new model to the same 5 strategies gives the coefficients shown in Table 3 with $R^2 = 0.924$

Using recursive feature elimination [10] it is possible to select the features (strategies) that give the best prediction for a given number of features. This *best* R^2 versus the number of features is shown in Figure 8.

Tables 4, 5 and 6 show the coefficients for linear models fitted to 5, 8 and 25 strategies with $R^2 = 0.924$, $R^2 = 0.967$ and $R^2 = 0.991$ respectively (8 strategies is the smallest number of strategies for which $R^2 > 96$ and 25 strategies is the smallest number of strategies for which $R^2 > 99$).

The predictions of these models are shown in Figure 9.

It is clear that the effectiveness of the predictive models with 5 strategies is low for the cluster of highly performing strategies (with a score greater than 2.5). To be able to obtain a good model even for high performing strategies 8 seem to provide a good predictive model.

Strategies	Coefficients	p -value	F -value
k69r	0.098640	0.000000	43.976036
k91r	0.207380	0.000000	56.667196
k40r	0.206440	0.000000	78.191831
k76r	0.059930	0.025992	5.207451
k67r	0.123740	0.000002	27.893148
Intercept	0.768410	NA	NA

Table 3: Linear model fitted to the same 5 strategies described in [2] with $R^2 = 0.885$

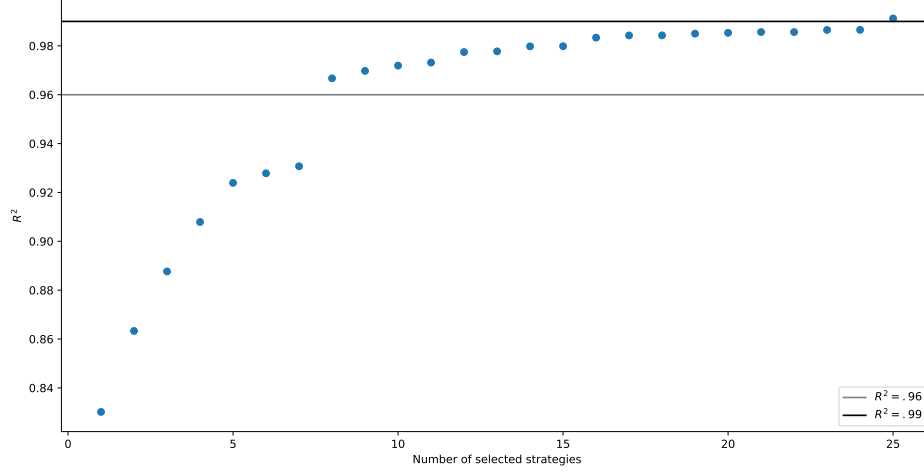


Figure 8: R^2 for models obtained using recursive feature elimination.

Strategies	Coefficients	p -value	F -value
k51r	0.458150	0.002438	10.001032
k56r	-24.586770	0.000000	48.729856
k59r	24.806190	0.000000	48.744926
k70r	0.261350	0.000000	79.384616
k89r	0.148550	0.015430	6.210811
Intercept	-0.399530	NA	NA

Table 4: Linear model best fitted to 5 strategies in the reproduced tournament with $R^2 = 0.924$

Strategies	Coefficients	p -value	F -value
k36r	0.133320	0.189994	1.756582
k51r	0.187270	0.002438	10.001032
k56r	-11.100380	0.000000	48.729856
k59r	11.231820	0.000000	48.744926
k62r	0.125620	0.703116	0.146621
k66r	0.151600	0.000000	135.614172
k70r	0.221780	0.000000	79.384616
k89r	0.172540	0.015430	6.210811
Intercept	-0.208980	NA	NA

Table 5: Linear model best fitted to 8 strategies in the reproduced tournament with $R^2 = 0.967$

Strategies	Coefficients	p -value	F -value
k32r	-0.060330	0.000000	186.247539
k34r	0.036030	0.000000	42.866846
k36r	0.044880	0.189994	1.756582
k41r	0.073800	0.000000	177.937911
k42r	0.044080	0.000000	79.351204
k43r	0.014170	0.000000	97.073236
k46r	0.044870	0.004805	8.566798
k47r	0.028380	0.000000	43.355694
k48r	0.029490	0.007426	7.670719
k51r	0.074000	0.002438	10.001032
k56r	-3.499880	0.000000	48.729856
k59r	3.551580	0.000000	48.744926
k60r	0.042590	0.000000	298.163936
k62r	0.042710	0.703116	0.146621
k66r	0.086580	0.000000	135.614172
k70r	0.067690	0.000000	79.384616
k73r	0.075500	0.000000	67.006588
k74r	0.023020	0.244116	1.383261
k75r	0.024290	0.169382	1.933926
k76r	0.031590	0.025992	5.207451
k81r	0.000130	0.000000	84.798727
k82r	0.032720	0.000001	29.385307
k89r	0.022020	0.015430	6.210811
k90r	0.041170	0.960298	0.002498
k91r	0.071910	0.000000	56.667196
Intercept	0.157940	NA	NA

Table 6: Linear model best fitted to 25 strategies in the reproduced tournament with $R^2 = 0.991$

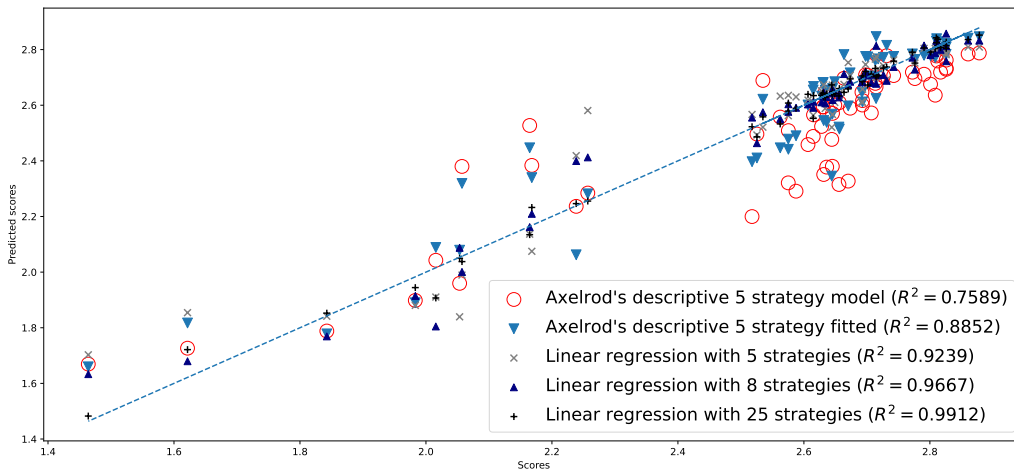


Figure 9: Predicting the performance of strategies using the 4 models discussed

This is the first known re-run of the tournament of [2] (which lead to [4], having over 33000 citations). Despite some misalignment with the results the overall principles are well-aligned. The original paper [2] used these results to give guidelines for effective behaviour:

1. Do not be envious by striving for a payoff larger than the opponent’s payoff.
2. Be “nice” by not being the first to defect.
3. Reciprocate both cooperation and defection; Be provokable to retaliation and forgiveness.
4. Do not be too clever by scheming to exploit the opponent.

Whilst these certainly apply to Axelrod’s particular tournament, in the next section we revisit the tournament and aim to see how correct the Game Theoretic community was to take this guidelines as a generalization.

4 Revisiting the tournament

In this section, the tournament of [2] will be expanded. Indeed, a large amount of research has gone on since Axelrod’s original work which include for example training of strategies using reinforcement learning [11] but also the discovery of Zero Determinant strategies [15]. This section aims to measure how well these strategies would have faired and **if** any of the original insights and conclusions would differ.

4.1 A large tournament

The Axelrod Python library has access to more than 200 strategies and regularly runs the largest iterated prisoners Dilemma tournament ever run (everytime a new strategy is contributed to the project the results are updated). We expand this tournament further by including all of the original Fortran strategies (omitting duplication of any strategies also in the Python library). This gives a tournament with 272 strategies. As for [7] the archival of all of the interactions and results from this large tournament are another contribution of this paper.

Table 7 shows the rankings of the top 20 strategies when including all strategies over 25,000 repetitions. The rank change across tournaments is shown in Figure 10. It is clear that a number of the mid table performing strategies move to the top ranking strategies in the more complex environment of the full tournament. The cooperation rate for the tournament that pits all the python library strategies against each other (without the original ones) is 0.619 . The overall cooperation rate of this larger tournament including the Fortran strategies is 0.626 , lower than the original tournament’s cooperation rate of 0.750 . The various cooperation rates of each strategy are shown in Figure 11.

	Mean Score	Rank	Coop. Rate	Original Rank	Reproduced Coop. Rate	Library Rank	Library Coop. Rate
EvolvedLookerUp2_2_2 [†]	2.802	1	0.756	NA	NA	1	0.736
Evolved HMM 5 [†]	2.788	2	0.751	NA	NA	2	0.742
Omega TFT: 3, 8	2.786	3	0.750	NA	NA	12	0.725
Evolved ANN 5 [†]	2.784	4	0.720	NA	NA	3	0.713
Evolved ANN [†]	2.782	5	0.739	NA	NA	5	0.727
Evolved FSM 16 [†]	2.780	6	0.731	NA	NA	4	0.713
Evolved FSM 16 Noise 05 [†]	2.776	7	0.726	NA	NA	6	0.717
PSO Gambler 2_2_2 [†]	2.760	8	0.702	NA	NA	7	0.692
Original Gradual	2.757	9	0.808	NA	NA	NA	NA
PSO Gambler 2_2_2 Noise 05 [†]	2.756	10	0.740	NA	NA	14	0.723
PSO Gambler Mem1 [†]	2.756	11	0.734	NA	NA	9	0.728
Evolved FSM 4 [†]	2.752	12	0.793	NA	NA	10	0.777
PSO Gambler 1_1_1 [†]	2.752	13	0.704	NA	NA	8	0.702
Gradual	2.746	14	0.763	NA	NA	15	0.793
DBS: 0.75, 3, 4, 3, 5	2.739	15	0.750	NA	NA	11	0.739
k42r	2.735	16	0.820	3	0.916	NA	NA
Winner12	2.733	17	0.688	NA	NA	16	0.682
Spiteful Tit For Tat	2.723	18	0.691	NA	NA	19	0.664
k60r	2.723	19	0.691	6	0.844	NA	NA
k85r	2.716	20	0.627	33	0.726	NA	NA
EugeneNier: (D,)	2.713	21	0.673	NA	NA	22	0.645
k80r	2.706	22	0.723	36	0.809	NA	NA
k32r	2.704	23	0.758	10	0.873	NA	NA
DoubleCrosser: (D, D)	2.703	24	0.693	NA	NA	13	0.677
k58r	2.703	25	0.746	21	0.852	NA	NA

Table 7: Top 25 strategies in the tournament when using all available strategies. Strategies denoted by a [†] have not been designed but instead trained using reinforcement learning as described in [11].

Figure 11 shows the distribution of cooperation rates. The reproduced cooperation rates are higher than the full tournament which are more closely aligned to the library tournament., Figure 13 shows the change of behaviour between tournaments. It is clear that the original fortran strategies cooperate less than they did in the original tournament.

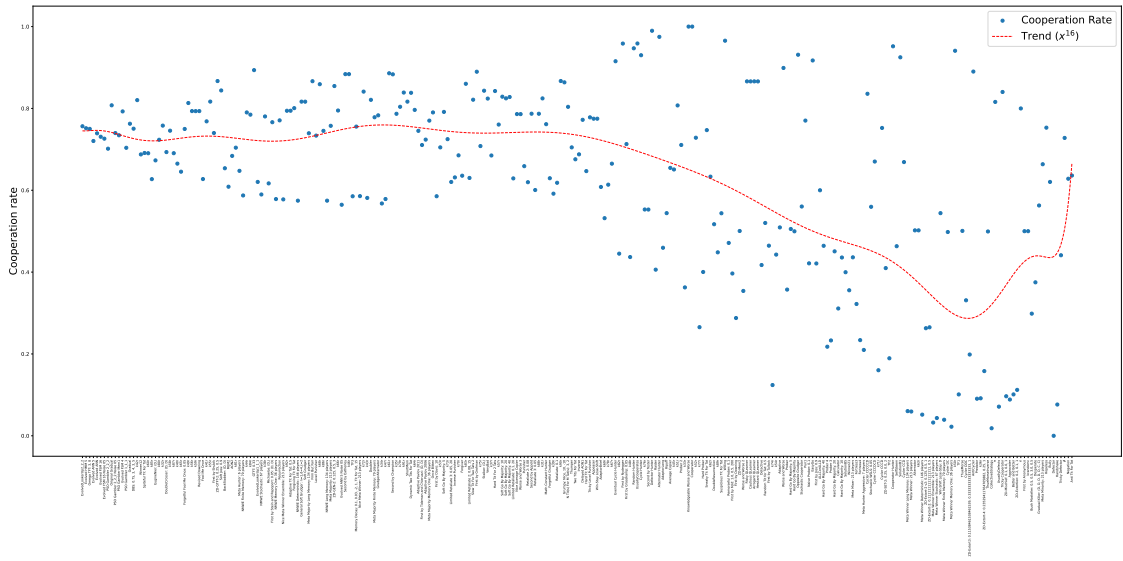


Figure 11: Cooperation rate versus rank for tournament with all available strategies

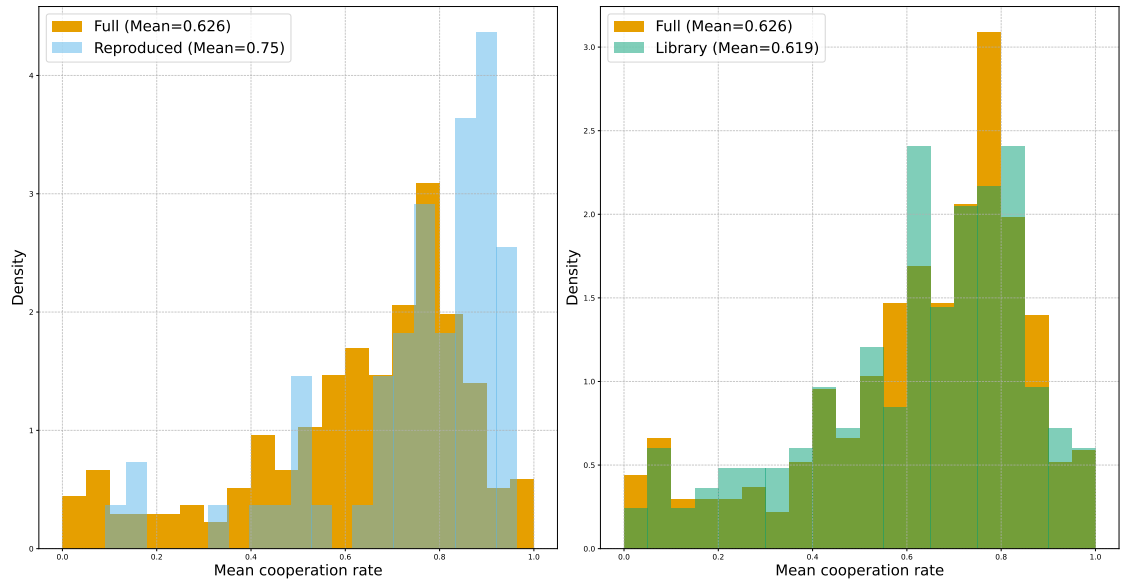


Figure 12: Distribution of cooperation rates for the full tournament.

Furthermore, looking at Table 7 the strategies that rank highly are sophisticated strategies that have been trained using reinforcement learning. The “Evolved” strategies have been trained using evolutionary algorithms and the “PSO” strategies have been trained using Particle Swarm Algorithms, these are described in [11]. Notably these high performing strategies do not necessarily follow the Axelrod’s guidelines. In fact they are more in line with the guidelines identified in [9] in which thousands of tournament variants identified the following key properties of successful strategies:

1. Be a little bit envious
2. Be “nice” in non-noisy environments or when game lengths are longer (if known)
3. Reciprocate both cooperation and defection appropriately; Be provokable in tournaments with short matches, and generous in tournaments with noise
4. It is ok to be clever
5. Adapt to the environment; Adjust to the mean population cooperation

The differences in the key properties are due at least partially to the higher proportion of cooperative strategies in the original tournament and the inclusion of more complex strategies from later research into the tournament presented above. TFT (k92r) still performs relatively well, as it naturally adjusts to the mean population cooperation rate (indeed for each match), but fails to exploit weaker opponents.

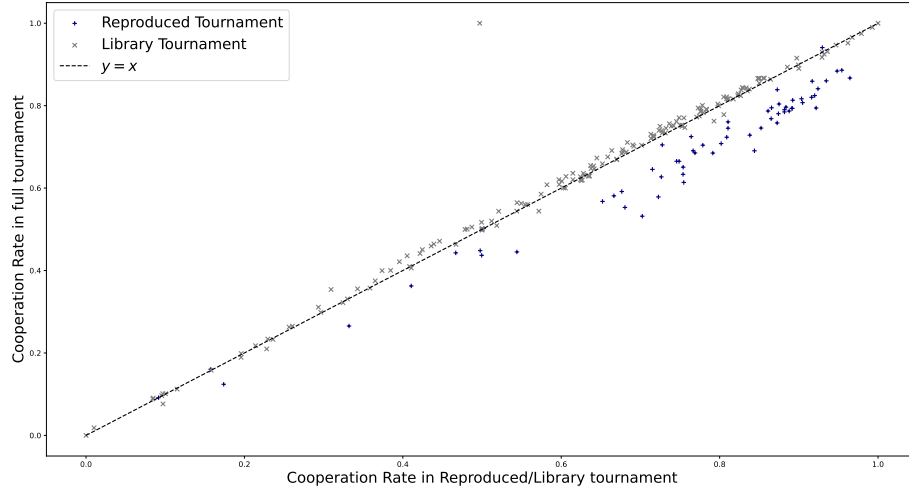


Figure 13: The relationship between cooperation rates in each tournament

In the next section we investigate if a small modification to Axelrod’s original tournament would have changed the conclusions. Specifically: would a small set of extra submissions changed the outcome?

4.2 Tournaments with extra invitations: Adding 1, 2, 3 and 4 strategies

Using the full tournament results it is possible to extract results for adding new strategies. The full tournament contains 272 strategies of which 63 are in the original tournament. This leads to a total of 78,760,605 new tournaments:

1. Adding 1 strategy gives: $\binom{209}{1} = 209$ tournaments.
2. Adding 2 strategy gives: $\binom{209}{2} = 21,736$ tournaments.
3. Adding 3 strategy gives: $\binom{209}{3} = 1,499,748$ tournaments.
4. Adding 4 strategy gives: $\binom{209}{4} = 77,238,876$ tournaments.

Name	Mean Score	Rank	Winner
Adaptive Tit For Tat: 0.5	2.876	2	k92r
GTFT: 0.33	2.846	3	k92r
Omega TFT: 3, 8	2.859	3	k92r
Firm But Fair	2.861	3	k92r
ZD-GTFT-2: 0.25, 0.5	2.868	3	k92r
Resurrection	2.867	3	k92r
Meta Majority: 213 players	2.860	3	k92r
Meta Majority Finite Memory: 79 players	2.856	3	k92r
Meta Majority Long Memory: 134 players	2.856	3	k92r
Soft Joss: 0.9	2.871	3	k92r
Forgiving Tit For Tat	2.854	3	k92r
Original Gradual	2.857	3	k92r
Meta Majority Memory One: 36 players	2.838	4	k92r
PSO Gambler 2_2_2 Noise 05 [†]	2.840	4	k92r
Gradual	2.831	7	k92r
First by Stein and Rapoport: 0.05: (D, D)	2.827	8	k92r
Spiteful Tit For Tat	2.826	8	k92r
GrudgerAlternator	2.825	8	k92r
EugeneNier: (D,)	2.815	10	k92r
First by Shubik	2.791	13	k92r

Table 8: Performance of extra strategy in Axelrod’s original tournament. Strategies denoted with a [†] were trained using reinforcement learning [11].

Table 8 shows the top ranking strategies when adding a single strategy to the original tournament.

Adding a single strategy from Table 8 does not lead to a winner from outside of the original strategies. Adding a single strategy either gives TFT, k42r (which ranked 2nd in the reproduced tournament and 3rd in the original tournament), or k60r (which ranked 9th in the reproduced tournament and 6th in the original tournament) as the winner however, adding 2 or 3 strategies does introduce new winners. Table 9 shows the proportion of times this happens. As more strategies are added it looks like the proportion of wins by k42r and k60r increase although this effect stops when adding 4 strategies. Figures 14 and 15 give the source code for these two strategies.

From inspection of the source code the behaviour of these strategies can be described as (these descriptions are taken from source code from [18] where these two strategies have been re-implemented in Python and verified to give accurate behaviours of the original Fortran code):

- k42r:

This player keeps track of the opponent’s responses to own behavior:

- `cd_count` counts: Opponent cooperates as response to player defecting.
- `cc_count` counts: Opponent cooperates as response to player cooperating.

The player has a defect mode and a normal mode. In defect mode, the player will always defect. In normal mode, the player obeys the following ranked rules:

1. If in the last three turns, both the player/opponent defected, then cooperate for a single turn.
2. If in the last three turns, the player/opponent acted differently from each other and they’re alternating, then change next defect to cooperate. (Doesn’t block third rule.)
3. Otherwise, do tit-for-tat.

Start in normal mode, but every 25 turns starting with the 27th turn, re-evaluate the mode. Enter defect mode if any of the following conditions hold:

- Detected random: Opponent cooperated 7-18 times since last mode evaluation (or start) AND less than 70% of opponent cooperation was in response to player’s cooperation, i.e. $\text{cc_count} / (\text{cc_count} + \text{cd_count}) < 0.7$

```

      FUNCTION K42R(JPICK,MOVEN,ISCORE,JSCORE,RANDOM,JA)
C BY OTTO BORUFSEN
C TYPED FROM FORTRAN BY AX, 1/25/79
      DIMENSION MHIST(2,2)
      K42R=JA      ! Added 7/27/93 to report own old value
C INITIALIZE FIRST MOVE
      IF(MOVEN.NE.1)GOTO 20
      L3MOV=0
      L3ECH=0
      IDEF=0
      ICOOP=0
      IPICK=0
      DO 10 I=1,2
      DO 10 J=1,2
10      MHIST(I,J)=0
      GO TO 500
20      IF(MOVEN.EQ.2)GOTO 25
C UPDATE MOVE HISTORY
      MHIST(I2PCK+1,JPICK+1)=MHIST(I2PCK+1,JPICK+1)+1
25      IF(IDEF.EQ.0)GOTO 30
C OPPONENT HAS BEEN PROVED "RANDOM" OR
C "DEFECTIVE",I DEFECT FOR 25 MOVES
      K42R=1
      GO TO 100
30      IF(IPICK.EQ.0.OR.JPICK.EQ.0)GOTO 40
C MUTUAL DEFECTIONS ON LAST MOVE.
      L3MOV=L3MOV+1
      IF(L3MOV.LT.3)GOTO 50
C MUTUAL DEFECTION ON
C LAST THREE MOVES.
C I COOPERATE ONCE ON NEXT MOVE.
      K42R=0
      L3MOV=0
      L3ECH=0
      GO TO 100
C ONE (OR BOTH) COOPERATED ON LAST MOVE.
40      L3MOV=0
      IF(IPICK.EQ.JPICK)GOTO 45
      IF(JPICK.NE.I2PCK.OR.IPICK.NE.J2PCK)GOTO 45
C ECHO-EFFECT ON LAST MOVE.
      L3ECH=L3ECH+1
      IF(L3ECH.LT.3)GOTO 50
C ECHO-EFFECT ON LAST THREE MOVES.
C MY NEXT DEFECTION WILL BE SUBSTITUTED
C BY A COOPERATION.
      L3ECH=0
      L3MOV=0
      ICOOP=1
      GO TO 50
45      L3ECH=0
C PLAY 'TIT FOR TAT' AS MAIN RULE.
50      K42R=JPICK
100     IF(MOD(MOVEN-2,25).NE.0.OR.MOVEN.EQ.2)GOTO 650
C ON EVERY 25 MOVES:
C CHECK IF OPPONENT SEEMS TO BE
C 'RANDOM' OR 'DEFECTIVE'.
      IDEF=0
      JNCOP=MHIST(1,1)+MHIST(2,1)
C IS OPPONENT 'RANDOM'?
      IF(JNCOP.GT.17)GOTO 155
      IF(JNCOP.LT.8)GOTO 130
      IF(100*MHIST(1,1)/JNCOP.LT.70)IDEF=1
      GO TO 155
C IS OPPONENT 'DEFECTIVE'?
130     IF(JNCOP.LT.3)IDEF=1
155     DO 160 I=1,2
      DO 160 J=1,2
160     MHIST(I,J)=0
      IF(IDEF.EQ.0)GOTO 650
C OPPONENT SEEMS TO BE
C 'RANDOM' OR 'DEFECTIVE'.
C I DEFECT FOR NEXT 25 MOVES.
      ICOOP=0
      L3MOV=0
      L3ECH=0
      GO TO 600
C I COOPERATE.
500     K42R=0
      GO TO 650
C I DEFECT.
600     K42R=1
650     IF(ICOOP.EQ.0.OR.K42R.EQ.0)GOTO 660
      ICOOP=0
      K42R=0
660     I2PCK=IPICK
      J2PCK=JPICK
      IPICK=K42R
      RETURN
      END

```

Figure 14: Fortran Source code for original k42r strategy submitted to Axelrod's second tournament.

```

      FUNCTION K6OR(J,M,K,L,R, JA)
C BY JIM GRAASKAMP AND KEN KATZEN
C FROM CARDS BY JM 2/22/79
      k6Or=ja      ! Added 7/27/93 to report own old value
      IF (M-1)1,1,2
1      ID=0
      K6OR=0
      GO TO 50
2      IF (ID-1)3,4,4
3      K6OR=J
      IF (M-11)50,5,6
5      IF (K-23)51,50,50
6      IF (M-21)50,7,8
7      IF (K-53)51,50,50
8      IF (M-31)50,9,10
9      IF (K-83)51,50,50
10     IF (M-41)50,11,12
11     IF (K-113)51,50,50
12     IF (M-51)50,13,14
13     IF (K-143)51,50,50
14     IF (M-101)50,15,50
15     IF (K-293)51,50,50
51     ID=1
4      K6OR=1
50     RETURN
      END

```

Figure 15: Fortran Source code for original k6or strategy submitted to Axelrod's second tournament.

- Detect defective: Opponent cooperated fewer than 3 times since last mode evaluation.

When switching to defect mode, defect immediately. The first two rules for normal mode require that the last three turns were in normal mode. When starting normal mode from defect mode, defect on first move.

- k60r: Play Tit-for-Tat at first, and track its own score. At select checkpoints, check for a high score. Switch to Default Mode if:
 - On move 11, score < 23
 - On move 21, score < 53
 - On move 31, score < 83
 - On move 41, score < 113
 - On move 51, score < 143
 - On move 101, score < 293

Once in Defect Mode, defect forever.

It is clear that both these strategies are more adaptable and "smarter" than Tit For Tat which is in line with the properties of high performing strategies of [9].

Winner	1 New (N = 209)	2 New (N = 21736)	3 New (N = 1499784)	4 New (N = 77238876)
Adaptive Tit For Tat: 0.5	NaN	0.000092	0.000232	0.000445
EugeneNier: (D,)	NaN	NaN	0.000001	0.000000
EvolvedLookerUp2_2_2†	NaN	NaN	NaN	0.000000
Firm But Fair	NaN	0.000046	0.000123	0.001323
First by Stein and Rapoport: 0.05: (D, D)	NaN	NaN	0.000001	0.000000
GTFT: 0.33	NaN	NaN	0.000020	0.000405
Gradual	NaN	NaN	0.000001	0.000002
Meta Majority Long Memory: 134 players	NaN	NaN	NaN	0.000000
Meta Majority: 213 players	NaN	NaN	NaN	0.000000
Omega TFT: 3, 8	NaN	0.000506	0.001457	0.002576
Original Gradual	NaN	0.000046	0.000218	0.002365
PSO Gambler 2_2_2 Noise 05†	NaN	0.000046	0.000227	0.000492
Resurrection	NaN	0.000092	0.000256	0.000267
Soft Joss: 0.9	NaN	0.000092	0.000285	0.005101
Spiteful Tit For Tat	NaN	NaN	0.000049	0.000139
ZD-GTFT-2: 0.25, 0.5	NaN	0.000138	0.000387	0.004531
k32r	NaN	NaN	0.000007	0.000789
k41r	NaN	NaN	0.000013	0.000000
k42r	0.148325	0.269415	0.366403	0.217232
k44r	NaN	0.000230	0.000568	0.004608
k49r	NaN	0.000138	0.000347	0.000231
k60r	0.004785	0.011180	0.018820	0.004514
k75r	NaN	0.000506	0.002448	0.000864
k92r	0.846890	0.717473	0.608137	0.754116

Table 9: Proportion of winners of the original tournament with 1, 2, 3 or 4 new strategies

4.3 Tournament with Zero Determinant Strategies

Since the work of [15] there has been much interest in Zero Determinant (ZD) strategies. In [17] a small tournament is presented pitting ZD strategies against a small collection of other strategies. Table 10 shows the rankings of the top 15 strategies when adding all the ZD strategies from [17] to Axelrod's second tournament. The change of rank across the tournaments is shown in Figure 16. The tournament exhibits limited relative change in performance, with the ZD strategies appearing evenly distributed throughout the rankings.

The overall cooperation rate of this tournament is 0.732 and the various cooperation rates are shown in Figure 17 (ordered by rank). The distribution is shown in Figure 18.

Figure 19 shows the relative cooperation rates and here it identifies that the Zero Determinant strategies in fact adapt their cooperation rate to the original tournament. They cooperate more than they do in their original tournament.

The final consideration is an external modification to the environment: what happens when there is noise: a non-zero probability of the chosen action by a strategy to be flipped.

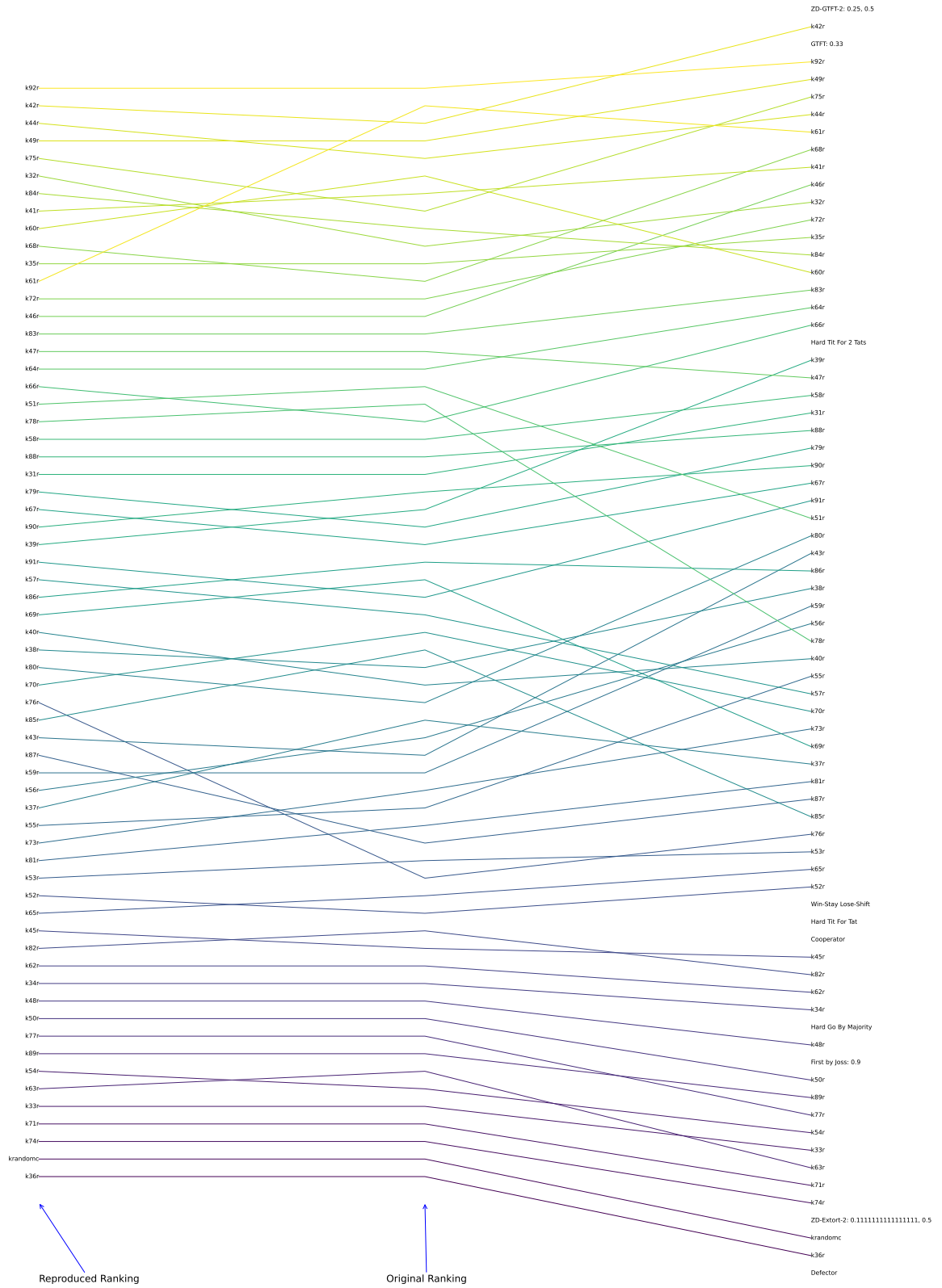


Figure 16: The change of rank in the tournament composed of the original strategies and the Zero Determinant strategies from [17]

Name	Mean Score	Rank	Mean Cooperation Rate	Reproduced Mean Cooperation Rate	S. and P. Mean Cooperation Rate
ZD-GTFT-2: 0.25, 0.5	2.834	1	0.926	NaN	0.855
k42r	2.825	2	0.903	0.916	NaN
GTFT: 0.33	2.819	3	0.941	NaN	0.893
k92r	2.812	4	0.888	0.922	0.731
k49r	2.787	5	0.879	0.892	NaN
k75r	2.786	6	0.782	0.802	NaN
k44r	2.781	7	0.866	0.882	NaN
k61r	2.773	8	0.945	0.954	NaN
k68r	2.768	9	0.904	0.917	NaN
k41r	2.761	10	0.847	0.865	NaN
k46r	2.758	11	0.939	0.948	NaN
k32r	2.758	12	0.849	0.873	NaN
k72r	2.756	13	0.915	0.924	NaN
k35r	2.747	14	0.863	0.888	NaN
k84r	2.745	15	0.861	0.882	NaN
k60r	2.735	16	0.814	0.844	NaN
k83r	2.725	17	0.922	0.935	NaN
k64r	2.710	18	0.876	0.884	NaN
k66r	2.702	19	0.861	0.866	NaN
Hard Tit For 2 Tats	2.701	20	0.926	NaN	0.807
k39r	2.700	21	0.671	0.676	NaN
k47r	2.696	22	0.709	0.723	NaN
k58r	2.694	23	0.846	0.852	NaN
k31r	2.690	24	0.908	0.920	NaN
k88r	2.684	25	0.851	0.861	NaN

Table 10: Top 15 strategies in the tournament composed of the original strategies and the Zero Determinant strategies from [17]

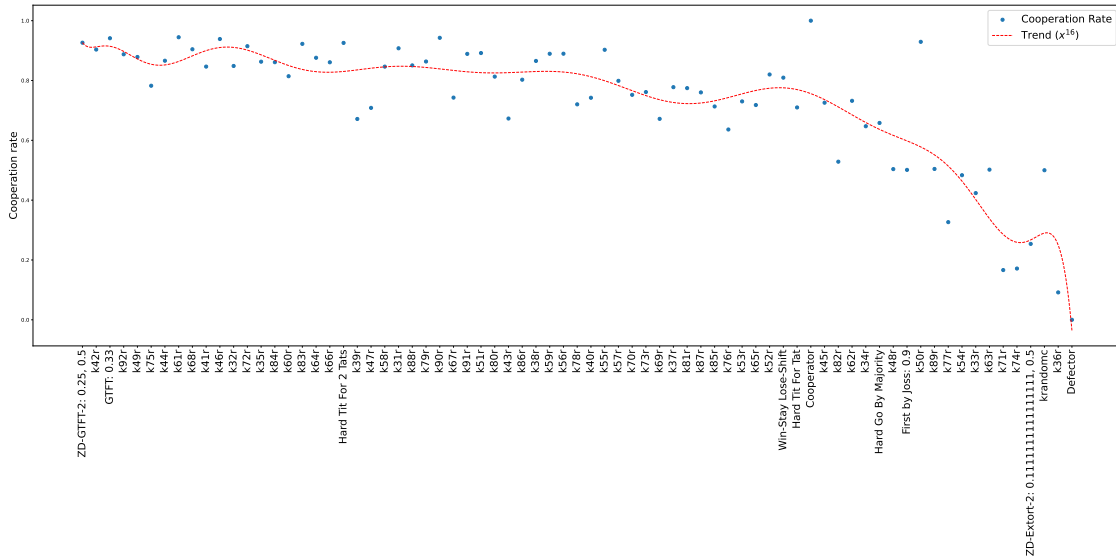


Figure 17: Cooperation rate versus rank for the Stewart and Poltkin tournament

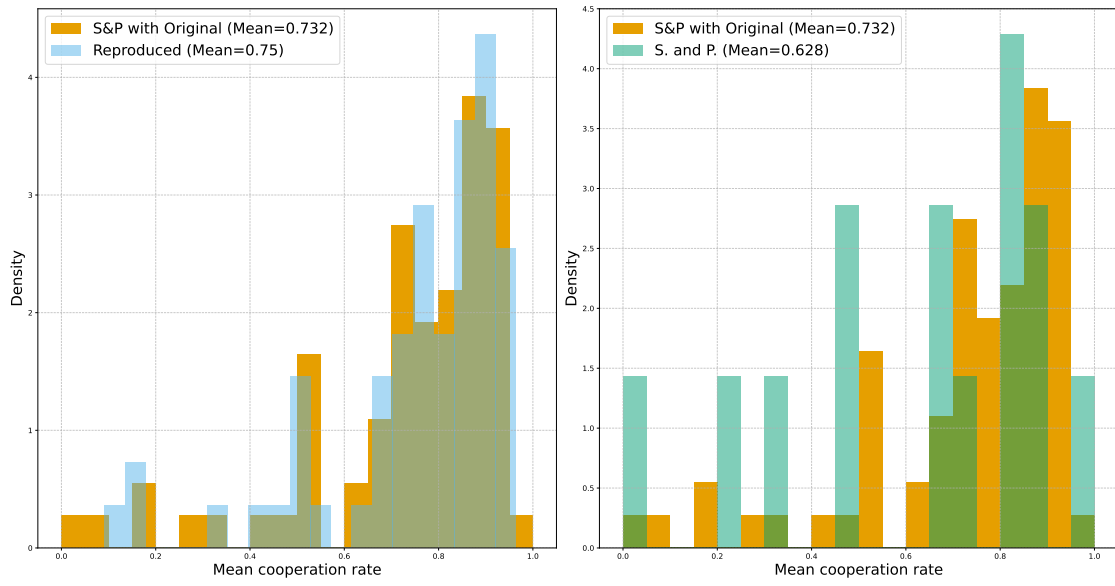


Figure 18: Distribution of cooperation rates for the Stewart and Plotkin tournament

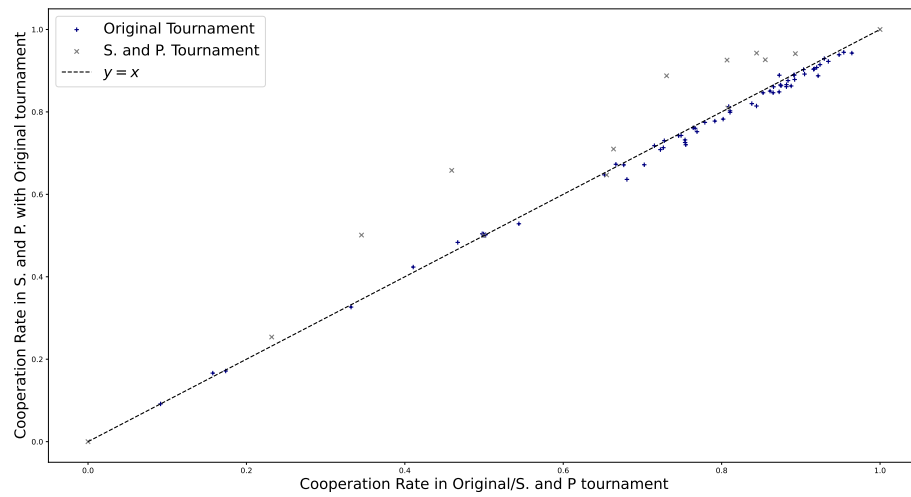


Figure 19: The relationship between cooperation rates in each tournament

4.4 Tournaments with noise

Tables 11, 12 and Figures 20 to 27 show the same set of results for two new runs of the full tournament but with either 1% or 5% chance of a change of chosen action. We once again see a number of the original mid performing strategies perform highly in noisy environments.

	Mean Score	Rank	Coop. Rate	Original Rank	Reproduced Coop. Rate	Library Rank	Library Coop. Rate
Evolved FSM 16 Noise 05 [†]	2.699	1	0.680	NA	NA	6	0.717
DBS: 0.75, 3, 4, 3, 5	2.670	2	0.731	NA	NA	11	0.739
Evolved ANN [†]	2.658	3	0.731	NA	NA	5	0.727
Omega TFT: 3, 8	2.652	4	0.682	NA	NA	12	0.725
k65r	2.651	5	0.611	47	0.715	NA	NA
k81r	2.648	6	0.634	43	0.779	NA	NA
Evolved ANN 5 [†]	2.642	7	0.671	NA	NA	3	0.713
BackStabber: (D, D)	2.641	8	0.654	NA	NA	18	0.641
PSO Gambler 2_2_2 Noise 05 [†]	2.640	9	0.647	NA	NA	14	0.723
k80r	2.625	10	0.722	36	0.809	NA	NA
DoubleCrosser: (D, D)	2.625	11	0.690	NA	NA	13	0.677
Evolved ANN 5 Noise 05 [†]	2.623	12	0.714	NA	NA	45	0.544
Evolved HMM 5 [†]	2.622	13	0.621	NA	NA	2	0.742
Fool Me Once	2.618	14	0.556	NA	NA	17	0.624
Evolved FSM 16 [†]	2.616	15	0.647	NA	NA	4	0.713
k47r	2.611	16	0.613	16	0.723	NA	NA
Gradual	2.608	17	0.723	NA	NA	15	0.793
k43r	2.607	18	0.643	39	0.667	NA	NA
Forgetful Fool Me Once: 0.05	2.606	19	0.686	NA	NA	20	0.753
Original Gradual	2.602	20	0.778	NA	NA	NA	NA
k56r	2.602	21	0.710	38	0.891	NA	NA
Revised Downing	2.601	22	0.710	NA	NA	NA	NA
k59r	2.601	23	0.710	40	0.892	NA	NA
Spiteful Tit For Tat	2.600	24	0.590	NA	NA	19	0.664
k42r	2.600	25	0.767	3	0.916	NA	NA

Table 11: Top 15 strategies in the full tournament with 1% noise. Strategies denoted with a [†] were trained using reinforcement learning [11].

	Mean Score	Rank	Coop. Rate	Original Rank	Reproduced Coop. Rate	Library Rank	Library Coop. Rate
DBS: 0.75, 3, 4, 3, 5	2.510	1	0.567	NA	NA	11	0.739
Evolved ANN 5 Noise 05 [†]	2.480	2	0.648	NA	NA	45	0.544
k80r	2.461	3	0.623	36	0.809	NA	NA
Evolved FSM 16 Noise 05 [†]	2.456	4	0.526	NA	NA	6	0.717
k47r	2.449	5	0.492	16	0.723	NA	NA
k43r	2.448	6	0.560	39	0.667	NA	NA
k38r	2.433	7	0.622	34	0.874	NA	NA
Evolved ANN [†]	2.430	8	0.577	NA	NA	5	0.727
BackStabber: (D, D)	2.426	9	0.458	NA	NA	18	0.641
Revised Downing	2.421	10	0.512	NA	NA	NA	NA
k59r	2.420	11	0.512	40	0.892	NA	NA
k56r	2.419	12	0.511	38	0.891	NA	NA
DoubleCrosser: (D, D)	2.415	13	0.529	NA	NA	13	0.677
k81r	2.406	14	0.469	43	0.779	NA	NA
k55r	2.403	15	0.528	42	0.903	NA	NA
Level Punisher	2.400	16	0.567	NA	NA	44	0.731
Omega TFT: 3, 8	2.399	17	0.502	NA	NA	12	0.725
k65r	2.397	18	0.387	47	0.715	NA	NA
k66r	2.389	19	0.679	20	0.866	NA	NA
k39r	2.378	20	0.619	25	0.676	NA	NA
k42r	2.376	21	0.637	3	0.916	NA	NA
Spiteful Tit For Tat	2.373	22	0.388	NA	NA	19	0.664
k49r	2.366	23	0.696	4	0.892	NA	NA
PSO Gambler 2_2_2 Noise 05 [†]	2.363	24	0.515	NA	NA	14	0.723
k60r	2.357	25	0.439	6	0.844	NA	NA

Table 12: Top 15 strategies in the full tournament with 5% noise. Strategies denoted with a [†] were trained using reinforcement learning [11].

The overall cooperation rates in this tournament go down, for 1% it is 0.597 and for 5% it is 0.51. This general trend is highlighted in the Figures where we see that the strategies that had a high cooperation rate were more likely to cooperate less. In this setting Tit For Tat does not rank in the top 15 strategies whereas trained strategies or indeed a sophisticated strategy specifically designed to take advantage of noise (the DBS strategy) perform well.

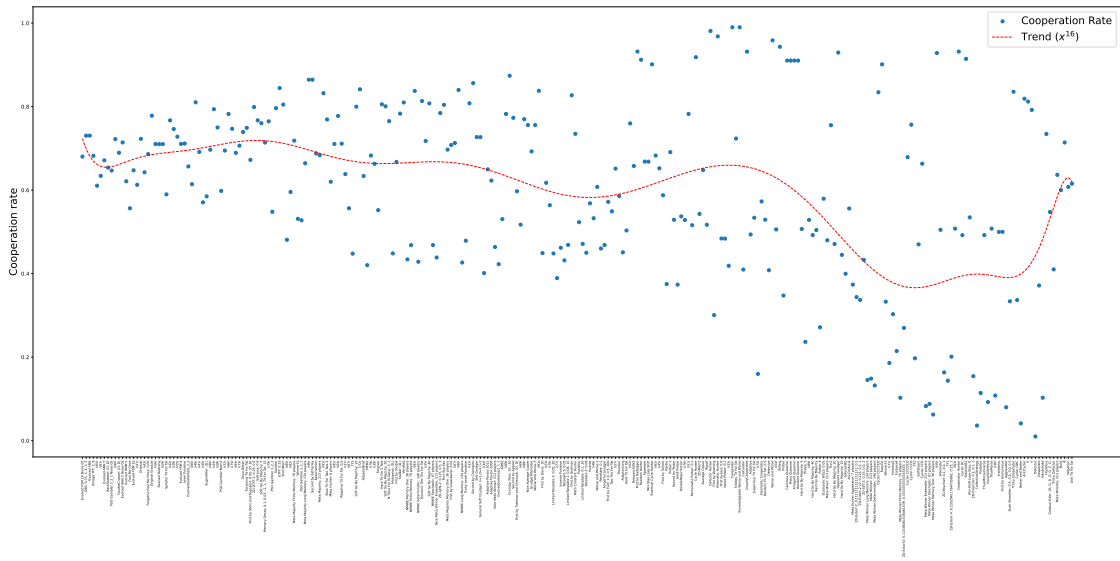


Figure 22: Cooperation rate versus rank for 1% noise

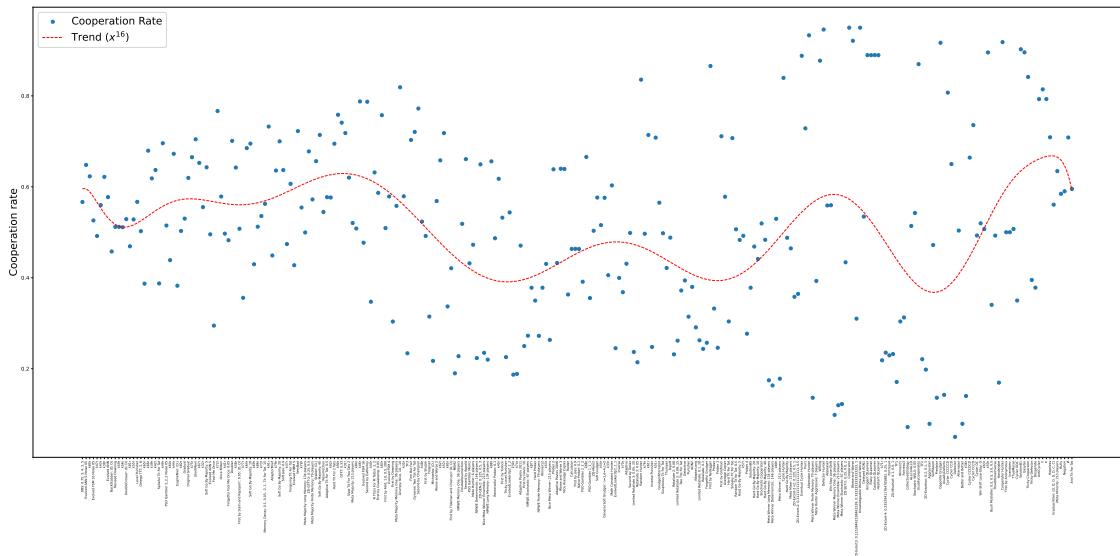


Figure 23: Cooperation rate versus rank for 5% noise

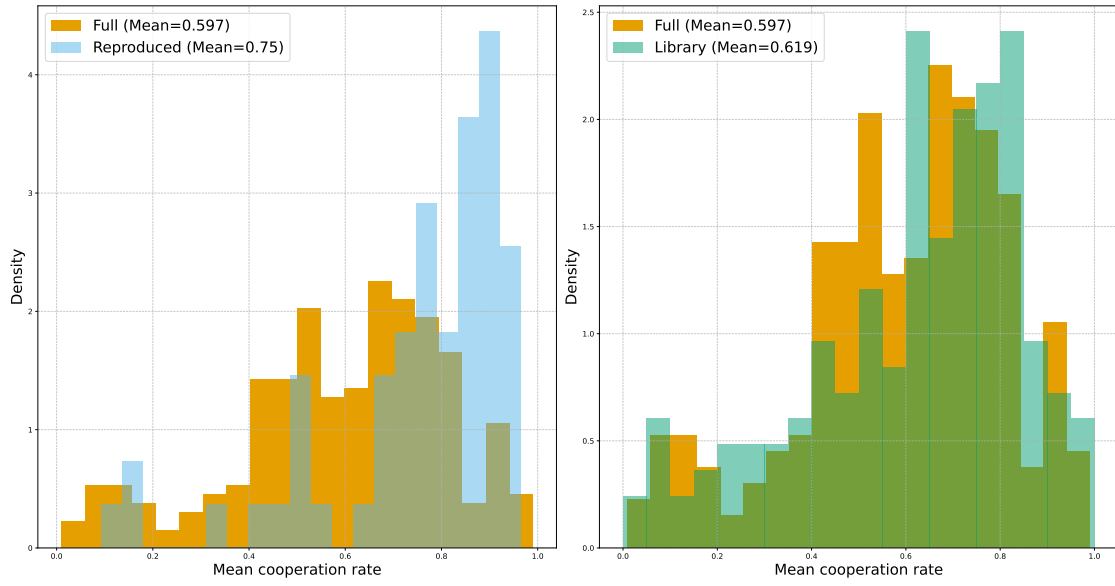


Figure 24: Distribution of cooperation rates for 1% noise

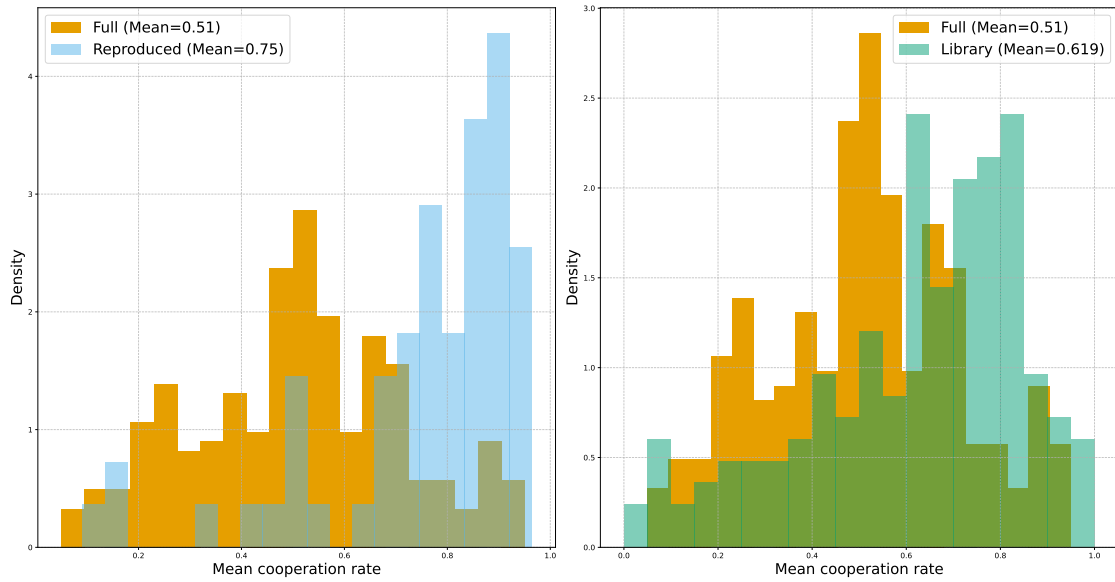


Figure 25: Distribution of cooperation rates for 5% noise

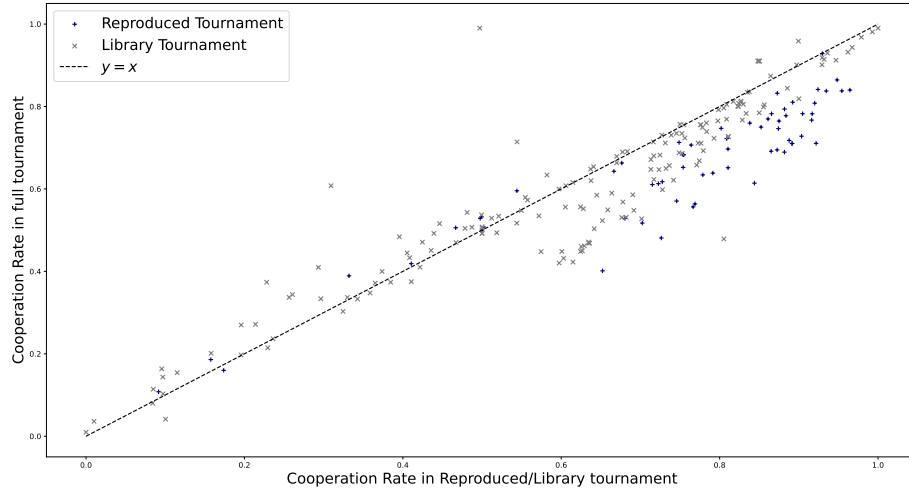


Figure 26: The relationship between cooperation rates in each tournament for 1% noise

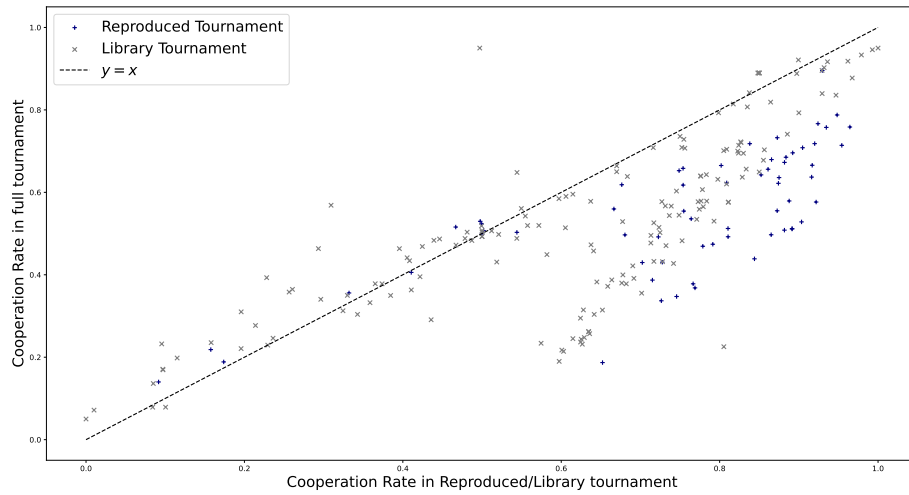


Figure 27: The relationship between cooperation rates in each tournament for 5% noise

5 Conclusion

The work in this paper is the first of its kind: we have packaged according to current best practices code that was written in the 1980s. The archive of this work available at ...has been written to high standards of best practice [19] and now benefits from a fully automated test suite. All modification to the original code files has been done in a systematic and reviewable approach which can be found at <https://github.com/Axelrod-Python/axelrod-fortran>. Using this, we have been able to re-run the original second Axelrod tournament, re-examine its outcomes, and investigate several variations. The reproduction matches well except for a single outlier strategy (Champion).

Axelrod’s work—highly cited and often treated as foundational—originates from just two specific tournaments. The widely accepted guidelines for effective cooperation stem directly from these early experiments. However, recent studies such as [9] have demonstrated that these rules are highly context-dependent and can fail to generalize well to broader settings.

The work presented in this paper further supports this view, showing that even small modifications to Axelrod’s original tournament setup could have significantly altered the conclusions drawn. By examining the impact of introducing just one or two additional strategies, we find that the overall interpretation of the results shifts: while a Tit For Tat-like strategy may still have emerged as the winner, it would have done so via a considerably more complex dynamic. Moreover, when we attempt to approximate the full strategy space using the largest available collection of strategies, the original submissions perform poorly. They are consistently outperformed by more sophisticated strategies—many of which arise through evolutionary processes rather than deliberate design.

Finally, the Axelrod Python maintainers have translated 28 of the original Fortran strategies to the Python programming language which is a somewhat archaeological exercise and benefits understanding and reproducibility of the work. Expanding this process in the open and reproducible framework offered by open source research software is an important exercise for future work.

Acknowledgements

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A List of original players

- | | |
|--|---|
| 1. k31r - Original rank: 23. Authored by Gail Grisell | 16. k46r - Original rank: 14. Authored by Graham J Eatherley |
| 2. k32r - Original rank: 10. Authored by Charles Kluepfel | 17. k47r - Original rank: 16. Authored by Richard Hufford |
| 3. k33r - Original rank: 59. Authored by Harold Rabbie | 18. k48r - Original rank: 53. Authored by George Hufford |
| 4. k34r - Original rank: 52. Authored by James W Friedman | 19. k49r - Original rank: 4. Authored by Rob Cave |
| 5. k35r - Original rank: 11. Authored by Abraham Getzler | 20. k50r - Original rank: 54. Authored by Rik Smoody |
| 6. k36r - Original rank: 63. Authored by Roger Hotz | 21. k51r - Original rank: 18. Authored by John William Colbert |
| 7. k37r - Original rank: 37. Authored by George Lefevre | 22. k52r - Original rank: 48. Authored by David A Smith |
| 8. k38r - Original rank: 34. Authored by Nelson Weiderman | 23. k53r - Original rank: 45. Authored by Henry Nussbacher |
| 9. k39r - Original rank: 25. Authored by Tom Almy | 24. k54r - Original rank: 58. Authored by William H Robertson |
| 10. k40r - Original rank: 35. Authored by Robert Adams | 25. k55r - Original rank: 42. Authored by Steve Newman |
| 11. k41r - Original rank: 7. Authored by Herb Weiner | 26. k56r - Original rank: 38. Authored by Stanley F Quayle |
| 12. k42r - Original rank: 3. Authored by Otto Borufsen | 27. k57r - Original rank: 31. Authored by Rudy Nydegger |
| 13. k43r - Original rank: 39. Authored by R D Anderson | 28. k58r - Original rank: 21. Authored by Glen Rowsam |
| 14. k44r - Original rank: 5. Authored by William Adams | 29. k59r - Original rank: 40. Authored by Leslie Downing |
| 15. k45r - Original rank: 50. Authored by Michael F McGurrin | 30. k60r - Original rank: 6. Authored by Jim Graaskamp and Ken Katzen |

31. k61r - Original rank: 2. Authored by Danny C Champion
32. k62r - Original rank: 51. Authored by Howard R Hollander
33. k63r - Original rank: 57. Authored by George Duisman
34. k64r - Original rank: 17. Authored by Brian Yamachi
35. k65r - Original rank: 47. Authored by Mark F Batell
36. k66r - Original rank: 20. Authored by Ray Mikkelsen
37. k67r - Original rank: 27. Authored by Craig Feathers
38. k68r - Original rank: 12. Authored by Francois Leyvraz
39. k69r - Original rank: 29. Authored by Johann Joss
40. k70r - Original rank: 32. Authored by Robert Pebly
41. k71r - Original rank: 60. Authored by James E Hall
42. k72r - Original rank: 13. Authored by Edward C White Jr
43. k73r - Original rank: 41. Authored by George Zimmerman
44. k74r - Original rank: 61. Authored by Edward Friedland
45. k75r - Original rank: 8. Authored by Paul D Harrington
46. k76r - Original rank: 46. Authored by David Gladstein
47. k77r - Original rank: 55. Authored by Scott Feld
48. k78r - Original rank: 19. Authored by Fred Mauk
49. k79r - Original rank: 26. Authored by Dennis Ambuehl and Kevin Hickey
50. k80r - Original rank: 36. Authored by Robyn M Dawes and Mark Batell
51. k81r - Original rank: 43. Authored by Martyn Jones
52. k82r - Original rank: 49. Authored by Robert A Leyland
53. k83r - Original rank: 15. Authored by Paul E Black
54. k84r - Original rank: 9. Authored by T Nicolaus Tideman and Paula Chieruzz
55. k85r - Original rank: 33. Authored by Robert B Falk and James M Langsted
56. k86r - Original rank: 28. Authored by Bernard Grofman
57. k87r - Original rank: 44. Authored by E E H Schurmann
58. k88r - Original rank: 22. Authored by Scott Appold
59. k89r - Original rank: 56. Authored by Gene Snodgrass
60. k90r - Original rank: 24. Authored by John Maynard Smith
61. k91r - Original rank: 30. Authored by Jonathan Pinkley
62. k92r - Original rank: 1. Authored by Anatol Rapoport
63. krandomc - Original rank: 62. Authored by None